Inspiring Excellence

CSE421: Computer Networks
Project Report
Project Title: "Topic 9 -> End of the Experiment"

Group No: 2505, **CSE421 Lab Section:** 25, **Fall 2025**	
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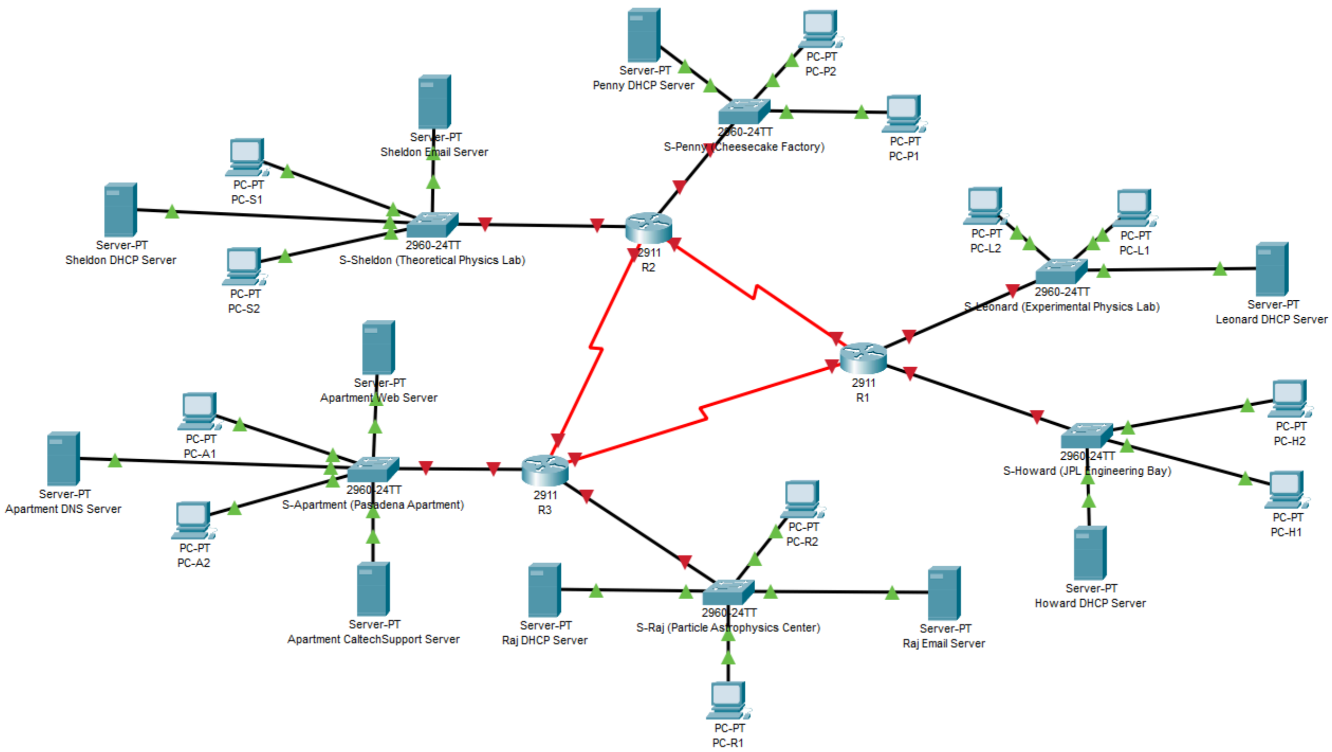
1.Network Topology Diagram:



2.Configuration Commands of all the Routers:

R1:

```
enable
configure terminal
interface g0/0
ip address 10.0.4.1 255.255.254.0
no shutdown
exit
interface g0/1
ip address 10.0.8.1 255.255.255.0
no shutdown
exit
interface s0/0/0
ip address 10.0.9.97 255.255.255.252
clock rate 64000
no shutdown
exit
interface s0/0/1
ip address 10.0.9.105 255.255.255.252
clock rate 64000
no shutdown
```

R2:

```
enable
configure terminal
interface g0/0
ip address 10.0.0.1 255.255.252.0
no shutdown
exit
interface g0/1
ip address 10.0.9.1 255.255.255.192
no shutdown
exit
interface s0/0/0
ip address 10.0.9.101 255.255.255.252
clock rate 64000
no shutdown
exit
interface s0/0/1
ip address 10.0.9.106 255.255.255.252
clock rate 64000
no shutdown
```

R3:

```
enable
configure terminal
interface g0/0
ip address 10.0.6.1 255.255.254.0
no shutdown
exit
interface g0/1
ip address 10.0.9.65 255.255.255.224
no shutdown
exit
interface s0/0/0
ip address 10.0.9.98 255.255.255.252
clock rate 64000
no shutdown
exit
interface s0/0/1
ip address 10.0.9.102 255.255.255.252
clock rate 64000
no shutdown
```

We will use:

- OSPF ONLY between R1 and R3
- R2 (Sheldon + Penny) uses PURE STATIC routing
- Leonard is the REQUIRED TRANSIT for the floating route

This is the only clean way to satisfy all routing and connectivity constraints.

R1:

```
configure terminal
router ospf 1
network 10.0.4.0 0.0.1.255 area 0
network 10.0.8.0 0.0.0.255 area 0
network 10.0.9.96 0.0.0.3 area 0
exit
```

R3:

```
configure terminal
router ospf 1
network 10.0.6.0 0.0.1.255 area 0
network 10.0.9.64 0.0.0.31 area 0
network 10.0.9.96 0.0.0.3 area 0
exit
```

R2 (Static Routes on R2):

```
enable
configure terminal

! To Leonard (R1)
ip route 10.0.4.0 255.255.254.0 10.0.9.105

! To Howard (R1)
ip route 10.0.8.0 255.255.255.0 10.0.9.105

! To Raj (R3)
ip route 10.0.6.0 255.255.254.0 10.0.9.102

! To Apartment (R3)
ip route 10.0.9.64 255.255.255.224 10.0.9.102

exit
```

R1 (Static Routes on R1):

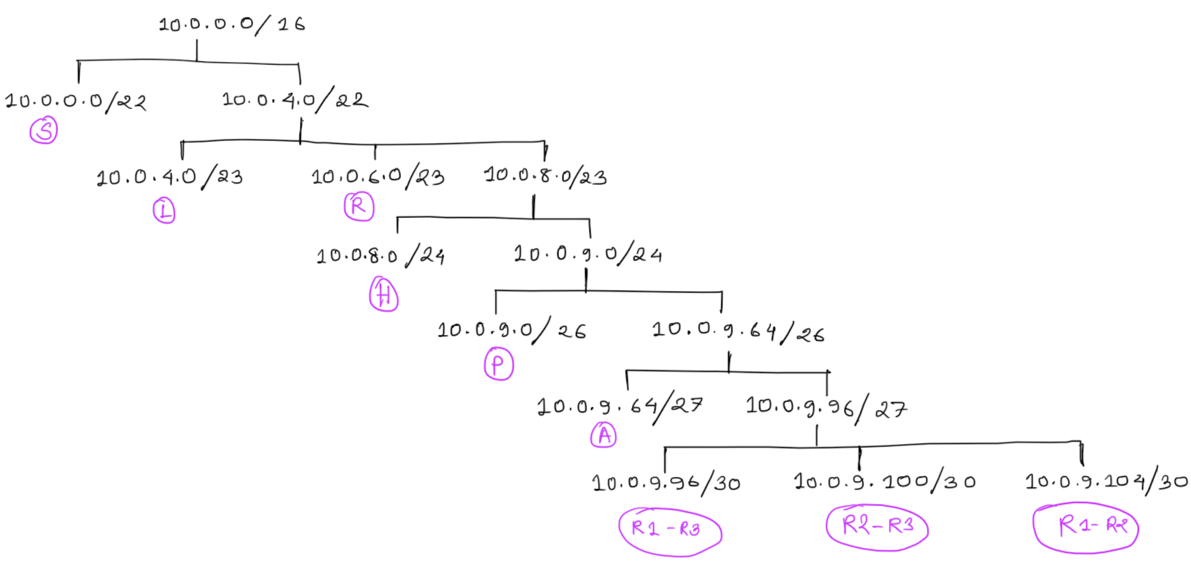
```
enable
configure terminal

! To Sheldon (R2)
ip route 10.0.0.0 255.255.252.0 10.0.9.106

! To Penny (R2)
ip route 10.0.9.0 255.255.255.192 10.0.9.106

exit
```

R3 (Static Routes on R3):

```
enable
configure terminal

! To Sheldon
ip route 10.0.0.0 255.255.252.0 10.0.9.101

! To Penny
ip route 10.0.9.0 255.255.255.192 10.0.9.101

! Floating Static Route to Raj from Apartment via Leonard, AD = 25
ip route 10.0.6.0 255.255.254.0 10.0.9.97 25

exit
```

Router	Zone	Routing Type
R1	Leonard + Howard	OSPF + Static
R2	Sheldon + Penny	Static
R3	Raj + Apartment	OSPF + Static + Floating

3. VLSM Tree and Calculations:



Here, we tried subnetting a base network address “10.0.0.0 /16”.

We assumed that the given host number excluded the router of that network. So we added 2 with the given host number as one for the “main” character (as mentioned in the PDF) and one for the router as a part of that network. In our topology, there were 9 networks in total.

Furthermore, the apartment had 21 members to be connected with the other zones. Here, we assumed the DNS, DHCP, and the CaltechSupport server were excluded from the 21 hosts, which means those 21 members were personal devices as end users. With the router counted, a total of 25 hosts were there for the apartment network.

Network Name	Hosts	Prefix Mask, n	Octet, k = ceil(n/8)	Increment, 2^(8k-n)	Network Address	First Address	Last Address	Broadcast Address
Theoretical Physics Lab (Sheldon)	814	/22	3	4	10.0.0.0 /22	10.0.0.1 /22	10.0.3.254 /22	10.0.3.255 /22
Experimental Physics Lab (Leonard)	362	/23	3	2	10.0.4.0 /23	10.0.4.1 /23	10.0.5.254 /23	10.0.5.255 /23
Particle Astrophysics Center (Raj)	277	/23	3	2	10.0.6.0 /23	10.0.6.1 /23	10.0.7.254 /23	10.0.7.255 /23
JPL Engineering Bay (Howard)	192	/24	3	1	10.0.8.0 /24	10.0.8.1 /24	10.0.8.254 /24	10.0.8.255 /24
Cheesecake Factory Back Office (Penny)	60	/26	4	64	10.0.9.0 /26	10.0.9.1 /26	10.0.9.62 /26	10.0.9.63 /26
Pasadena Apartment	25	/27	4	32	10.0.9.64 /27	10.0.9.65 /27	10.0.9.94 /27	10.0.9.95 /27
R1 <-> R3	2	/30	4	4	10.0.9.96 /30	10.0.9.97 /30	10.0.9.98 /30	10.0.9.99 /30
R2 <-> R3	2	/30	4	4	10.0.9.100 /30	10.0.9.101 /30	10.0.9.102 /30	10.0.9.103 /30
R1 <-> R2	2	/30	4	4	10.0.9.104 /30	10.0.9.105 /30	10.0.9.106 /30	10.0.9.107 /30

4. IP Address Table: